

■ E-Commerce Data Analysis

■■■:	Dier
■■■■■■■:	19 ■■■■
■■■■■■■■■■■■■■■:	Data Analyst / Junior
■■■■■■■■■■■■■■■:	International Digital University

■ September 01, 2026

Python (Pandas)

1. Aggregating Revenue Data by Product Category

```
df.agg({'revenue': ['min', 'mean', 'max']})
```

```
min: 11.21 | mean: 1021.96 | max: 4119.33
```

Electronics: 11.21 | mean: 1021.96 | max: 4119.33 | Clothing: 11.21 | mean: 1021.96 | max: 4119.33 | Home: 11.21 | mean: 1021.96 | max: 4119.33 | Beauty: 11.21 | mean: 1021.96 | max: 4119.33

2. Grouping and Sorting Revenue Data

```
df.groupby('product_category')['revenue'].sum().sort_values(ascending=False)
```

```
Electronics: 1.8M | Clothing: 1.5M | Home: 982K | Beauty: 766K
```

Electronics — 37% of total revenue. Electronics is the highest revenue category.

3. Applying a Custom Function to Delivery Data

```
df['deliver_type'] = df['delivery_days'].apply(lambda x: 'fast' if x <= 5 else 'slow')
```

```
df.groupby('deliver_type')['quantity'].sum()
```

fast: 10,000 | slow: 10,000

4. Grouping and Summing Quantity Data

```
df.groupby('product_category')['quantity'].sum()
```

```
Electronics: 7,109 | Clothing: 6,171 | Home: 3,949 | Beauty: 2,995 (Total: 20,224)
```

Electronics is the highest quantity category.

5. Grouping and Summing Revenue Data by Region

```
df.groupby('region')['revenue'].sum().sort_values(ascending=False)
```

```
West: 1.3M | North: 1.2M | South: 1.2M | East: 1.2M
```

West is the highest revenue region (~2-3% of total revenue).

SQL (SQLite/PostgreSQL)

1. SQL 101: Basic Syntax and Query Structure

Example: `SELECT * FROM users WHERE age > 25;` West: 1.3M (1,289) | North: 1.2M (1,276) | South: 1.2M (1,208) | East: 1.2M (1,227)

GROUP BY ORDER BY

2. SQL 101: Advanced Query Techniques (VIP)

Example: `SELECT * FROM users WHERE ID > 1890;` ID 1890: 12,529 | ID 1779: 9,201 | ID 1608: 8,767 | ID 1907: 6,392

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3. SQL 101: Data Manipulation and Aggregation

Example: `SELECT * FROM users WHERE Card = 1;` Card: 2,270 (56%) | COD: 1,774 (44%) | Wallet: 956 (24%)

COD

4. SQL 101: Data Analysis and Reporting (VIP)

Example: `SELECT * FROM users WHERE Clothing = 1;` Clothing: 18.43% | Beauty: 18.14% | Home: 17.83% | Electronics: 17.67%

Clothing. ~18% = 6-7K

5. SQL 101: Data Analysis and Reporting

Example: `SELECT * FROM users WHERE Clothing = 1;` Clothing: +8,596% | Electronics: +6,514% | Home: +5,364% | Beauty: +3,754%

= Clothing 86

1■ Electronics — ■■■■■■■■ ■■■■■■■■■■

[illegible]

2008-09-07

[illegible][illegible]

10% = 0.10 = 10/100 = 1/10. 1/10 = 0.10.

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2-3% ██████████ ████████ ████████████████. ████████████████ ████████ ████████, █████ ████████████████████.

5 ■■■■■■■■■■ — ■■■■■■■■■■

56% ■■■■, 44% COD. ■■■■ ■■■■■■ ■■■■■■ — ■■■■■■■■■■ ■■■■■■ ■■■■■■.

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■ ■■■■■■ ■■■■■■■■	5.8M ■■■■■■■■
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■ ■■■■■■■■■■ ■■■■■■	5 actionable insights

3,391 ■■■■■■

□ □ □

5.8M 

■ ■ ■

Python, SQL, Excel, Power BI

■ ■ ■

5 actionable insights

